

MD ZUNAYEED KAMAL

Cell: (+1)260-579-6920 Email: zunayeed.edu@gmail.com

CAREER SUMMARY

- Telecommunications and IT professional with 14 years of overall experience, including 6 years of hands-on 5G NR and 4G LTE RAN experience in commissioning, integration, feature validation, optimization, automation, and end-to-end service delivery.
- 5G RAN engineer with experience supporting 5G NSA and SA deployments, including gNB configuration, CU/DU concepts, NR/LTE interworking, carrier aggregation, dual connectivity, mobility, radio-resource management, and feature validation.
- Practical expertise in 5G NR protocol analysis and troubleshooting across SDAP, PDCP, RLC, MAC, RRC, PHY interactions, QoS, bearer management, and UE radio-access capability validation.
- Experienced in 5G UE/device interoperability and performance testing, including registration, attach, data-session validation, mobility testing, VoNR/IMS validation, log and trace analysis and end-to-end root-cause investigation.
- Strong working knowledge of 3GPP 5G NR specifications and 5G service requirements for eMBB, URLLC, and mMTC, with focus on throughput, latency, reliability, coverage, capacity, and scheduling performance.
- Proficient in Python-based automation for test execution, log parsing, regression testing, KPI analysis, reporting, and RAN validation efficiency improvements.
- Working knowledge of 5G Core architecture and RAN-to-core integration, including AMF, SMF, UPF, PCF, UDM, AUSF, NRF, NSSF, NEF, AF, service-based interfaces, HTTP/2, REST APIs, and JSON.
- Analytical, detail-oriented engineer with strong communication, presentation, documentation, and cross-functional collaboration skills; experienced delivering projects from requirements and design through integration, testing, optimization, and production deployment.

TECHNICAL SKILLS

RAN Technologies	5G NR (SA/NSA), 4G LTE, gNodeB/eNodeB, AirScale, RAN commissioning & integration
PHY Layer	MCS tables (64QAM/256QAM), DMRS Type 1/2, NR-ARFCN, BandwidthCombinationSet, PDSCH/PUSCH scheduling
NR-RAN Protocols	RLC, PDCP, RRC, SDAP, 3GPP Release 15/16/17 specifications
5G Core & Interfaces	NSSF, NEF, NRF, UDM, PCF, AF, AUSF, AMF, SMF; HTTP/2, REST API, JSON in 5G
Testing	UE interoperability (IOT) testing, mobility/handover, VoLTE, VoNR, Dynamic Spectrum Sharing, Network Slicing, multi-UE throughput testing
KPI & Optimization	KPI analysis, SW/HW release validation, performance optimization, troubleshooting
Tools	QXDM, Wireshark, Iperf, Tcpdump, Component Trace, Emil
Automation & Scripting	Python 3.11+, automation scripting for test execution, log collection, reporting
Languages	Python 3.11+, Java, SQL, C++, C, JavaScript
Technologies	J2EE, JDBC, JPA, JSP, Servlets, JMS, Microservices, Spring Boot, Log4j
Markup Languages	HTML, CSS, XML, jQuery, AngularJS, Ajax
ORM Technology	Hibernate
Servers	Tomcat, JBoss, Jetty

Application Frameworks	Flask, Spring 5.0, Spring Core, Spring Boot, Spring IO, Spring Data, Bootstrap
Databases	SQL Server, Oracle, PostgreSQL, pgAdmin, MySQL
Operating Systems	Windows, Red Hat Linux 9, Ubuntu
IDEs	Cursor, Eclipse, STS, PyCharm, Visual Studio Code
Version Control / Build Tools	Git
Testing & CI/CD	JUnit, Mockito, GitHub Actions
Cloud	Amazon AWS, Pivotal Cloud Foundry (PCF)

WORK EXPERIENCE

F2G Solutions Inc, Plano, Texas, Sub Contract - Nokia, Corporate Headquarters, Coppell, Texas, USA)

January 2021 – Present

System Engineer (Interoperability Testing & Technical Support – On Contract)

Texas, USA

5G NR & 4G LTE Commissioning & Integration

- Commissioning and integration of Nokia's advanced wireless products such as LTE network elements, gNodeB, eNodeB, AirScale, radio units covering SA and NSA configurations
- Building test lines for NR and LTE sites and modifying site commissioning files with parameters aligned to customer configuration and IPs for OAM, CU, and DU planes with network interfaces S1, X2, NG, and XP using XML.
- Performing SW/HW release validations on customer-specific configurations in the lab and OTA for NR/LTE – TDD & FDD.
- Testing and validating Nokia's NR and LTE RAN features and functionalities with respect to 3GPP, and performing KPI analysis.
- Supporting performance and optimization, performing SW updates, and troubleshooting with Nokia 5G NR and LTE RAN.

5G Use Case Demonstration

- Leveraging Nokia's expertise to provide a platform to demonstrate, optimize, and validate diverse 5G-centric use cases and over-the-top applications.
- Realizing advanced 5G features in the product development cycle through proofs of concept, trials, demos, and system testing with real-world applications, characterizing 5G network and device performance and KPIs.
- Creating commercial videos to showcase Nokia's new radio units and RAN features, demonstrating maximum data reception and high throughput capabilities.

Automation

- Executing automation scripts for test case execution, log collection and uploads, test attribute reporting, and Pass/Fail calculation, eliminating manual repetitive work for both Delivery and KPI test runs.
- Automating the SCF file to get relevant info for testing as per customer requests.

5G Test Cases and Device Testing

- Testing UE devices from Nokia, Samsung, Qualcomm, Apple, for interoperability and compatibility across features including mobility, handover, VoLTE, VoNR, Dynamic Spectrum Sharing, and Network Slicing.
- Performing multi-UE tests to achieve high data throughput.
- Verified a VoNR call using a Qualcomm MTP (X105, SDX90) and Nokia's gNB 5G26R3 SW build, and a VoLTE call using an MTP with Nokia's eNB and gNB, achieving a successful NSA call and an SA call with up to 6 component carriers (6CC).
- Successfully performed FR1-FR2 and FR1-FR1 handovers using an MTP and Nokia's gNB 5G26R2 SW build with an NSA/SA call.
- Operated network monitoring software and packet analyzers including Wireshark, Iperf, Tcpdump, BIP log analysis, and Component Trace to test wireless network performance of Nokia's eNodeB and

gNB networks, analyzing and optimizing UE functionality.

Custom Magnetics Inc.

February 2018 – December 2020

Quality Control Engineer

Fort Wayne, IN

- Designed custom magnetic components to meet electrical, mechanical, thermal, regulatory, cost, and manufacturability requirements.
- Provided engineering support for customers, manufacturing, and cross-functional teams to improve product quality, cost, and production efficiency.
- Led migration of M2M transformer test programs from ODI to Voltech AT5600; developed, debugged, and validated automated test programs through sanity and unit testing.
- Troubleshoot transformer test-system and design issues, supporting reliable product validation and production testing.

Purdue University

May 2017 – April 2018

Teaching Assistant

Fort Wayne, Indiana, USA

- Instructed lab classes on C and C++ and software demonstration classes; assisted students with their projects.
- Assisted the professor in preparing course homework, checking homework, midterm and final exams, and proctoring exams.

SARBS Communication Limited

June 2015 – July 2016

Account Manager – Service & Solutions (Technical)

Bangladesh

- Gathered and analyzed customer requirements to support technical solution design, workflow development, and system enhancements.
- Managed local and international client relationships, including technical coordination, requirement follow-up, and service support.

Vision Telecom Limited

June 2012 – July 2014

Network Engineer – Transmission & Core Network

Bangladesh

- Configured and supported Tejas 16XX-series multiplexers and STM transmission links based on network design, capacity, and operational requirements and troubleshoot transmission network issues
- Coordinated with internal teams, vendors, and customers on technical requirements, system changes, incident follow-up, and service support day-to-day transmission and core-network operations.

EDUCATION

Purdue University, Fort Wayne, Indiana, USA

Fall 2016 – Spring 2018

MS in Applied Computer Science

GPA: 3.82/4

North South University, Bangladesh

Summer 2012 – Spring 2014

Master of Business Administration (Major in Finance)

GPA: 3.67/4

North South University, Bangladesh

Summer 2007 – Spring 2011

BS in Electronics and Telecommunication Engineering

GPA: 3.70/4

CERTIFICATION

Oracle Certified Professional: Java SE 11 Developer (Part 1)

September 2019

Java programming, modular development, and secure coding practices.

PUBLICATION (CONFERENCE PAPER)

Md Zunayeed Kamal, Dr. Arshad M. Chowdhury, “A PAPR Reduction Approach in OFDM Optical-Wireless-Access Networks Using Blind Symbol Power Estimation,” International Conference on Connected Vehicles and Expo (ICCVE), Las Vegas, IEEE, USA, December 6, 2013.

ACADEMIC ACHIEVEMENTS

- Purdue University: Graduated with distinction; GPA: 3.82/4.00.
- North South University: Magna Cum Laude; received a 75% merit scholarship and lecturer-position offer in Electrical and Computer Engineering.

EXTRA-CURRICULAR

- Completed Cisco Certified Network Associate (CCNA) training, Parts 1–4 (December 2011 – June 2012), American International University, Dhaka, Bangladesh.

- Attended a workshop on Algorithmic, Mathematical, and Statistical Foundations of Data Science and Applications (April 12–13, 2019) at Purdue University, West Lafayette.
- Conference paper peer reviewer for IEEE Global Communications Conference, Atlanta, Georgia, USA, 2013.
- AlgoExpert Certificate of Completion — Completed 100 coding interview questions on AlgoExpert.
- Plays soccer regularly at Irving Community Center, Irving, Texas